Advanced Programming Practice · 21CSC203P

Unit 5 · Session 4 · SLO 1 — Solved Worksheet

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Unit â€“ 5 : Session â€“ 4 : SLO â€“ 1 | Symbolic Programming Concepts

I. Questions & Answers

1. Symbolic computation â€” computation in which mathematical objects (variables, expressions, equations) are manipulated as exact symbols rather than approximate numbers, e.g. solving xÂ² â^' 1 = 0 to get exactly x = 1 and x = â^'1.

2. Python library â€” SymPy is the standard Python library for symbolic computation (alternatives: SageMath; Octave/Mathematica outside Python).

3. Example of a symbolic equation â€” Eq(x**2 + 2*x + 1, 0), i.e. xÂ² + 2x + 1 = 0; SymPy solves it exactly: solve(x**2 + 2*x + 1, x) â†' [â^'1].